LOGIC AND PROOFS

Propositional Equivalences

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OUTLINE

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PROPOSITIONAL EQUIVALENCES

Terminology

Tautology: A compound proposition that is always true, no matter what the truth values of the propositional variables that occur in it, e.g. $p \vee \neg p$.

Contradiction: A compound proposition that is always false, e.g. $p \wedge \neg p$.

Contingency: A compound proposition that is neither a tautology nor a contradiction.

Examples of a Tautology and a Contradiction.

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

Logical Equivalences

1. Two compound propositions p and q are called logically equivalent if $p \leftrightarrow q$ is a tautology, i.e. the compound propositions p and q are equivalent if and only if the columns giving their truth values agree.
2. The notation $p \equiv q$ denotes that p and q are logically equivalent.
3. One way to determine whether two compound propositions are equivalent is to use a truth table.

Example: Show that (1) $\neg(p \vee q)$ and $\neg p \wedge \neg q$, and (2) $\neg(p \wedge q)$ and $\neg p \vee \neg q$, are logically equivalent.

De Morgan's Laws.
$\neg(p \wedge q) \equiv \neg p \vee \neg q$
$\neg(p \vee q) \equiv \neg p \wedge \neg q$

Truth Tables for $\neg(p \vee q)$ and $\neg p \wedge \neg q$.

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

Figure: Because the truth values of the compound propositions $\neg(p \vee q)$ and $\neg p \wedge \neg q$ agree for all possible combinations of the truth values of p and q, it follows that $\neg(p \vee q) \leftrightarrow \neg p \wedge \neg q$ is a tautology and that these compound propositions are logically equivalent.

Example: Show that $p \to q$ and $\neg p \vee q$ are logically equivalent. (This is known as the **conditional-disjunction equivalence**).

Example: Show that $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are logically equivalent. This is the **distributive law of disjunction over conjunction.**

A Demonstration That $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ Are Logically Equivalent.

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

Figure: Because the truth values of $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ agree, these compound propositions are logically equivalent.

KEY LOGICAL EQUIVALENCES

Whenever we are dealing with laws that include T, that means it's a tautology. If F, it's a contradiction.

Logical Equivalences.

Equivalence	*Name*
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws

Equivalence	*Name*
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

Logical Equivalences Involving Conditional Statements.
$p \to q \equiv \neg p \lor q$
$p \to q \equiv \neg q \to \neg p$
$p \lor q \equiv \neg p \to q$
$p \land q \equiv \neg(p \to \neg q)$
$\neg(p \to q) \equiv p \land \neg q$
$(p \to q) \land (p \to r) \equiv p \to (q \land r)$
$(p \to r) \land (q \to r) \equiv (p \lor q) \to r$
$(p \to q) \lor (p \to r) \equiv p \to (q \lor r)$
$(p \to r) \lor (q \to r) \equiv (p \land q) \to r$

Logical Equivalences Involving Biconditional Statements.
$p \leftrightarrow q \equiv (p \to q) \land (q \to p)$
$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$
$p \leftrightarrow q \equiv (p \land q) \lor (\neg p \land \neg q)$
$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

Using De Morgan's Laws

Use De Morgan's laws to express the negations of "Miguel has a cellphone and he has a laptop computer" and "Heather will go to the concert or Steve will go to the concert."

Solution

Let
p: "Miguel has a cellphone", and q: "Miguel has a laptop computer"
"Miguel has a cellphone and he has a laptop computer" can be represented by $p \wedge q$. By the first of De Morgan's laws, $\neg(p \wedge q)$ is equivalent to $\neg p \vee \neg q$. Consequently, we can express the negation of the original statement as "Miguel does not have a cellphone or he does not have a laptop computer"

Let
r: "Heather will go to the concert", and s: "Steve will go to the concert."
"Heather will go to the concert or Steve will go to the concert" can be represented by $r \vee s$. By the second of De Morgan's laws, $\neg r \vee s$ is equivalent to $\neg r \wedge \neg s$. Consequently, we can express the negation of our original statement as "Heather will not go to the concert and Steve will not go to the concert."

Constructing New Logical Equivalences

A proposition in a compound proposition can be replaced by a compound proposition that is logically equivalent to it without changing the truth value of the original compound proposition.

Show that

$\neg(p \to q)$ and $p \land \neg q$ are logically equivalent.

We want to use logical identities to establish new logical identities. We establish this equivalence by developing a series of logical equivalences, using the equivalences in the Tables above, starting with $\neg(p \to q)$ and ending with $p \land \neg q$.

$$\begin{aligned}
\neg(p \to q) &\equiv \neg(\neg p \lor q) && \text{by the conditional-disjunction equivalence} \\
&\equiv \neg(\neg p) \land \neg q && \text{by the second De Morgan's Law} \\
&\equiv p \land \neg q && \text{by double Negation Law}
\end{aligned}$$

Use logical equivalences to show that
$\neg(\neg p \vee q) \equiv \neg q \wedge p$

$$\begin{aligned} \neg(\neg p \vee q) &\equiv \neg\neg p \wedge \neg q && \text{by the 1st De Morgan's Law} \\ &\equiv p \wedge \neg q && \text{by double Negation Law} \\ &\equiv \neg q \wedge p && \text{by Commutative Law} \end{aligned}$$

Therefore, $\neg(\neg p \vee p) \equiv \neg q \wedge p$.

Show that
a. $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$
b. $(p \wedge q) \to p \vee q$ is a tautology by developing a series of logical equivalences.

a. $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$

$$
\begin{aligned}
\neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) && \text{by the 2nd De Morgan's Law} \\
&\equiv \neg p \wedge [\neg(\neg p) \vee \neg q] && \text{by the 1st De Morgan's Law} \\
&\equiv \neg p \wedge (p \vee \neg q) && \text{by the Double Negation Law} \\
&\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) && \text{by the 2nd Distributive Law} \\
&\equiv \mathbf{F} \vee (\neg p \wedge \neg q) && \text{because } \neg p \wedge p \equiv \mathbf{F} \\
&\equiv (\neg p \wedge \neg q) \vee \mathbf{F} && \text{by Commutative Law for disjunction} \\
&\equiv \neg p \wedge \neg q && \text{by the Identity Law for F}
\end{aligned}
$$

$(p \wedge q) \to (p \vee q)$ is a tautology by developing a series of logical equivalences.

$$\begin{aligned}
(p \wedge q) \to (p \vee q) &\equiv \neg(p \wedge q) \vee (p \vee q) && \text{by } p \to q \equiv \neg p \vee q \\
&\equiv (\neg p \vee \neg q) \vee (p \vee q) && \text{by the 1st De Morgan's Law} \\
&\equiv (p \vee \neg p) \vee (q \vee \neg q) && \text{by the Commutative and Associative} \\
& && \text{Laws for Disjunction} \\
&\equiv T \vee T && \text{by the 1st Negation Law} \\
&\equiv T && \text{by the Domination Law}
\end{aligned}$$

Prove or disprove that:

$$p \to (q \wedge p) \equiv p \to q$$

State all the logical equivalences used at each step.

$$\begin{aligned}
p \to (q \wedge p) &\equiv (\neg p \vee (q \wedge p)) && \text{by } p \to q \equiv \neg p \vee q \\
&\equiv (\neg p \vee q) \wedge (\neg p \vee p) && \text{by the 1st Distributive law} \\
&\equiv (\neg p \vee q) \wedge T && \text{by the 2nd Negation law} \\
&\equiv (\neg p \vee q) && \text{by the Domination Law} \\
&\equiv p \to q && \text{by } p \to q = \neg p \vee q
\end{aligned}$$

Prove or disprove that:

$$(\neg q \wedge (p \to q)) \to \neg p \equiv T$$

State all the logical equivalences used at each step.

$$\begin{aligned}
(\neg q \wedge (p \to q)) \to \neg p &\equiv (\neg q \wedge (\neg p \vee q)) \to \neg p && \text{by } p \to q \equiv \neg p \vee q \\
&\equiv (\neg q \wedge \neg p) \vee (\neg q \wedge q) \to \neg p && \text{by the 1st Distributive law} \\
&\equiv (\neg q \wedge \neg p) \vee F \to \neg p && \text{by the 2nd Negation law} \\
&\equiv (\neg q \wedge \neg p) \to \neg p && \text{by the 2nd Identity Law} \\
&\equiv \neg(\neg q \wedge \neg p) \vee \neg p && \text{by } p \to q = \neg p \vee q \\
&\equiv (q \vee p) \vee \neg p && \text{by De Morgan's 1st law} \\
&\equiv (\neg p \vee p) \vee q && \text{by Commutative law and Associative lav} \\
&\equiv T \vee q && \text{by the 1st Negation law} \\
&\equiv T && \text{by the 1st Domination law}
\end{aligned}$$

The End